Automated Leukemia Detection based on White Blood Cell Localization and Multitask Learning

Abhinaya B[1], Ankita I B[1], Madhumithan S[1], Mahesh R[1], Dr Poongodi T[1*] and Dr. Natarajan Venkateswaran[1]

[1] Computer Science Engineering, School of Engineering, Dayananda Sagar University, Bengaluru, India.

Abstract: Leukemia identification from images of peripheral blood smears is very important for clinical decision-making, but manual microscopic analysis is time-consuming and subjective to errors. Existing work either relies on CNN-based classifiers that operate on pre-segmented images without morphological context, or work only on single-task transformer models that decouple blast detection from leukemia subtype classification limiting their diagnostic completeness. This paper proposes an automated, end-to-end leukemia blast cell detection system. Unlike prior works that treat detection and classification separately, this is the first integrated YOLOv11-based WBC localization with Vision Transformer multitask learning. A YOLOv11 object detection model is used to accurately identify and crop individual white blood cells (WBCs) from microscopic images. The cropped cells are then analyzed by a multitask network based on a vision transformer (ViT-Base-Patch16-224) model for simultaneous blast vs. non-blast cell and leukemia type identification and predict WBC cell attributes. By learning multiple related tasks together in a shared representation space, the proposed method can effectively learn discriminative morphological characteristics and improve generalization performance. The proposed system is tested on publicly available peripheral blood smear image datasets TXL-PBC dataset and WBCATT dataset based on metrics for the evaluation of classification performance. The results show that the proposed work can attain a classification accuracy of up to 98%, mAP@0.50 of 98% and mAP@0.50:0.95 of 87% for blood cell detection, and an overall F1-score of 98%.

Keywords: Leukemia detection, Peripheral blood smear analysis, YOLOv11, Vision Transformer, ViT-Base-Patch16-224, Multitask learning, WBC classification.

1 INTRODUCTION

Leukemia is a type of blood cancer that affects the blood stream and the bone marrow's blood production function by the excess production of immature or premature WBCs, commonly referred to as blast cells [1]. The blast cells cause the inefficiency of blood cell production, which may lead to conditions such as anemia, infections, and bleeding or bruising [2]. It is important to note that the early diagnosis of leukemia, improves the treatment and outcome. The most used technique for diagnosing patients with leukemia is done through observing the blood smear samples under the microscope, where expert hematopathologists evaluate the cell morphologies and identify any abnormalities found in them, as shown in Figure 1 [3]. In this diagnosis, the size and shape of nucleus, cytoplasmic properties are some important morphologies taken into consideration [4].

*Author for correspondence

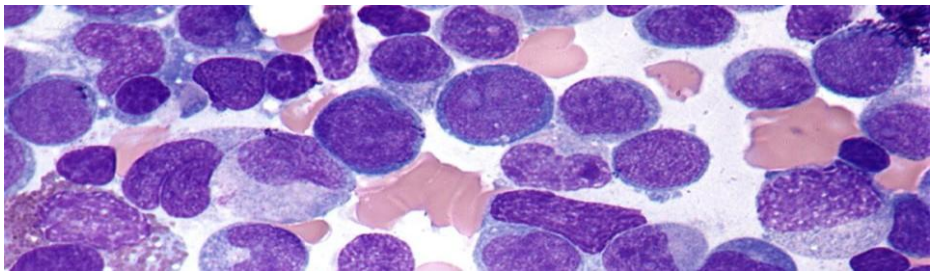


Figure 1. Sample peripheral blood smear image

Although manual diagnosis is quite accurate and common, it is time-consuming, requires skilled professionals, and is prone to variability, especially when the irregularities are not very prominent. Leukemia affects more than 470,000 people annually and is responsible for more than 300,000 deaths worldwide [5]. This illustrates the severity of this disease for global health. Recent developments in blood smear image analysis have advanced WBC classification and leukemia diagnosis [6-7]. However, most of the recent work done in the field of CNN-based approaches have demonstrated strong performance on pre-segmented, single-cell images; however, they are constrained by limited receptive fields that prevent reasoning over cellular morphology. This area of problem development emphasizes the need for an automated framework for classification and analysis of WBCs [8]. This technical gap - the absence of an end-to-end system that couples WBC localization with multitask transformer-based blast detection and subtype prediction is what this work directly addresses.

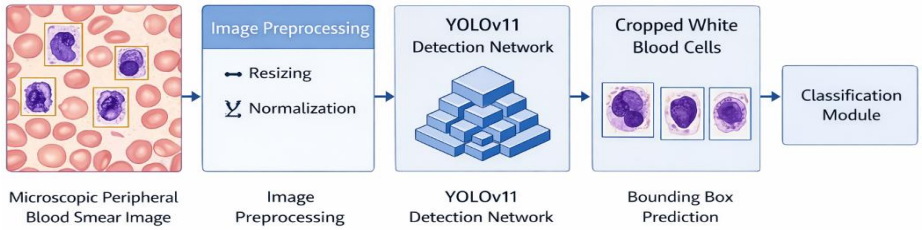


Figure 2. Yolo v11 WBC localization workflow

Object detection models enable automatic localization of WBCs in full microscopic blood smear images, as illustrated in Figure. 2. YOLO-based detectors provide a good tradeoff between accuracy, speed and automates cell detection without human intervention [9].



Figure 3. Vision Transformer–based multitask learning architecture

Recently, Vision Transformers (ViTs) have shown strong capability in modeling complex visual patterns using self- attention which can capture long-range

dependencies across the whole image, as shown in Figure 3 [10]. Additionally, multitask learning further improves diagnostic performance by jointly learning related classification tasks within a shared representation space [20]. In the diagnosis of leukemia, the detection of blast cell and the type of leukemia are related to each other because both are based on the observation of the cells. Based on this concept, this work presents a unified system for the detection of leukemia that uses YOLO to detect WBCs and vision transformer to perform multiple tasks simultaneously, this work is the first to integrate YOLOv11-based WBC localization with ViT multitask learning for simultaneous blast detection, leukemia subtype classification, and morphological attribute prediction in a single end-to-end framework.

2 RELATED WORK

A comparative study of recent blood smear image analysis methods along with highlighting their key contributions and limitations are mentioned in Table 1.

Table 1. Comparative Table of Recent Leukemia Detection Techniques

Ref. No.	Reference	Method Used	Key Contribution	Limitation
[1]	Alomar et al. (2025)	Swin Transformer + Momentum Contrast (MoCo) contrastive learning	Built a hierarchical attention-based model using Swin Transformer feature extraction with self-supervised contrastive learning [19] for ALL subtype classification	Focuses solely on ALL subtype classification; does not integrate cell localization or multitask learning for simultaneous blast detection
[2]	Tanwar et al. (2025)	ResViT — hybrid ResNet-50 + dual-stream Vision Transformer	Combined CNN local feature extraction with transformer global attention	lacks end-to-end localization from full blood smear and does not perform multitask learning
[3]	Acevedo et al. (2020)	Morphological Attribute Annotation (WBC-ATT)	Provided a densely annotated dataset with 11 morphological attributes for Explainable AI (XAI)	Focused primarily on normal cell categories; lacks specific labels for all leukemia subtypes

[4]	Tan et al. (2022)	TBXPCB Multi-focus Dataset	Introduced a comprehensive dataset using multi-focus image stacks to improve classification accuracy	The multi-focus approach requires more storage and slower image acquisition hardware
[5]	Abbas et al. (2022)	GAN + CNN Transfer Learning	Used GANs to generate synthetic images, overcoming data imbalance	Synthetic images may lack the fine textures needed for clinical accuracy

Current approaches for leukemia detection are based on single-task models that only perform the classification task or use pre-segmented images without localization, which limits the completeness of diagnosis and causes redundant feature learning. Our multitask framework addresses this gap by jointly localizing WBCs and detecting and classifying the blasts. Medical image analysis has also used segmentation architectures such as U-Net [13] for extracting relevant regions from biomedical images. Extensions such as UNet++ [14] have further explored nested encoder-decoder architectures for improved medical image segmentation. A broader review of deep learning-based image segmentation methods highlights the importance of segmentation techniques in biomedical image analysis [15]. Transformer-based segmentation approaches such as UNETR [18] have also demonstrated the utility of transformer architectures to medical image segmentation.

3 PROPOSED METHODOLOGY

3.1 Overall System Workflow

1. The classification of blood cells and localization of WBCs from blood smear samples using a YOLOv11 object detection model.
2. For subsequent classification of WBC types and attribute prediction, a ViT-Base-Patch 16-224 version transformer is used.

This end-to-end automation design provides high diagnostic accuracy while excluding the manual diagnosis.

3.2 YOLO-Based White Blood Cell Localization

In the initial phase, from a blood smear image individual blood cells are classified and WBCs are localized. This process is challenging as there may be frequent overlapping of cells, stains may not be uniform and there might be background noises, making it hard to rely on manual techniques. In order to overcome this more efficiently, direct localization of WBCs from a whole blood smear image is done using an object detection model YOLOv11. YOLOv11 was chosen over YOLOv5 and YOLOv8 [16] due to its enhanced C3k2 bottleneck architecture and optimized anchor-free detection,

delivering superior performance in handling overlapping cells and complex staining patterns, for robust WBC localization.

$$\boldsymbol{B = \{(x_i, y_i, w_i, h_i)\}_{i=1}^{N}} \quad \text{Eq 1}$$

Let us denote the input image by I. We define (xi,yi) as the location of the detected cell's center, and (wi,hi) – its width and height respectively. The RBC count is decreased (anemia) due to reduced production of normal erythrocytes in the bone marrow. The WBC count is increased (leukocytosis) because of the number of blasts that are present in the circulation. The platelet count is decreased (thrombocytopenia) due to suppression of their production in the bone marrow.

3.3 Multitask Learning Architecture

The model uses a shared ViT-Base-Patch16-224 backbone to extract features, which are subsequently passed to two task-specific heads, one for blast vs. non-blast detection and another for leukemia subtype classification. The shared morphological characteristics help the simultaneous learning of the two tasks, avoiding the redundant feature learning as in Fig. 3. To solve the data imbalance issue, we use Focal Loss with class weighting that down-weights easy examples and penalizes hard negatives, thus enabling the model to concentrate on hard misclassifications and learn robust decision boundaries.

3.4 ViT-Base-Patch16-224 Feature Extraction

Each WBC image is then resized to 224x224 pixels and normalized before being input into the ViT-Base-Patch16-224 model. The image is then divided into 16x16 pixel patches, which are flattened into embedded vectors and combined with positional embedded vectors. These vectors are then passed through layers of a transformer encoder, which allows the model to understand morphological information of WBC cells such as nuclear morphology, chromatin, and cytoplasm. The ViT-Base-Patch16-224 backbone consists of 12 transformer encoder layers. Each encoding layer has an embedding dimension of 768 and 12 self-attention heads, giving rise to approximately 86 million trainable parameters. With the Adam optimizer [12], performed training for 50 epochs with a batch size of 32, an initial learning rate of 1×10^{-4} and a cosine annealing learning rate scheduler. Validation F1-score was applied for early stopping with a patience of 10 epochs to prevent overfitting. The complete end-to-end workflow of the proposed automated blast diagnostic system is shown in Figure 4.

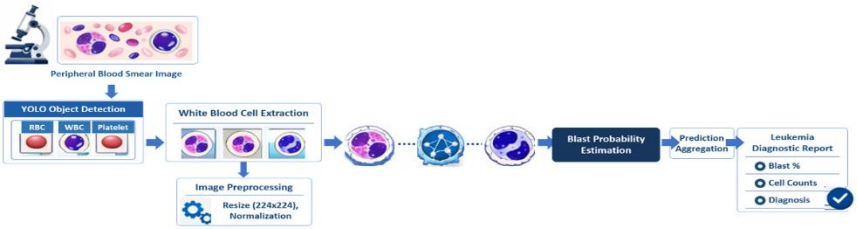


Figure 4. End-to-end workflow for automated blast diagnostics

3.5 Blast Cell Detection and Diagnosis

The multitasking vision transformer model predicts the blast probability for each detected WBC based on their morphological features. Blast-like cells are defined as those with two or more predicted morphological attributes exhibiting blast-like characteristics at a confidence score of >50%. This threshold was selected after a sensitivity analysis with three candidate values, which showed that 50% provides the best trade-off between reducing false negatives (more dangerous in a clinical scenario of leukemia screening) and maintaining an acceptable level of precision. We empirically optimized the 50% confidence threshold for blast-like cell identification by sensitivity analysis to balance minimization of false negatives (clinically critical in leukemia screening) with acceptable false positive rates, enabling reliable blast detection without causing diagnostic errors. The percentage of blasts is calculated as the ratio of number of blasts detected to the total number of WBCs detected . The diagnostic report is generated as follows :

$$\boldsymbol{Blast\ Percentage} = \frac{\boldsymbol{N_{Blast}}}{\boldsymbol{N_{WBC}}} \times \mathbf{100} \quad \text{Eq 2}$$

4 DATASET AND PREPROCESSING

4.1 Dataset Description

1) The TXL-PBC dataset, which was used for the detection stage, was formed by combining images from four sources to address the issues of annotation inconsistency and data scarcity. These sources include the BCCD database, which contains approximately 12,500 annotated blood smear images, BCDD – a peripheral smear dataset, PBC – a dataset of peripheral blood smear images annotated with labels for WBC classification, and Raabin-WBC – a large-scale dataset containing around 17,000 WBC images annotated with subtyping labels. To form the final detection set, all bounding boxes were converted to three classes of RBC, WBC, and Platelet, as they appeared in most databases with minor modifications to the labels. Additionally, annotations in different formats were converted to a standard YOLO format and duplicate images found using perceptual hashing were removed. The four databases differ in the staining procedure, magnification level, and hardware used to capture images, which introduces domain shift.

2) WBCAtt dataset is employed for the detailed analysis of WBCs. The dataset contains 10,298 images of individual cells. These images represent all five major types of WBCs.
a) Class Distribution: Neutrophils: 3,329 images, Eosinophils: 3,117 images, Monocytes: 1,420 images, Lymphocytes: 1,214 images, Basophils: 1,218 images.
b) Attributes: Each image is annotated with 11 different morphological features: Cell Size (2 classes): big, small, Cell Shape (2 classes): irregular, round, Nucleus Shape (6 classes): irregular, segmented-bilobed, segmented-multilobed, unsegmented-band, unsegmented-indented, unsegmented-round, Nuclthereear-to-Cytoplasmic Ratio (N:C

Ratio) (2 classes): high, low, Chromatin Density (2 classes): densely, loosely, Cytoplasm Vacuoles (2 classes): present, absent, Cytoplasm Texture (2 classes): clear, frosted, Cytoplasm Color (3 classes): blue, light blue, purple blue, Granule Type (4 classes): coarse, absent, round, small, Granule Color (4 classes): absent, pink, purple, red, Granularity (2 classes): present, absent

This yields more than 113,000 attribute labels, which are used to train the interpretable attribute branch of the ViT model. Data augmentation techniques can improve training data diversity and model robustness in deep learning applications [17].

5 RESULTS AND ANALYSIS

5.1 White Blood Cell Detection Results

Using the standard object detection metrics, the TXL - PBC dataset is used to train YOLOv11 model for WBC localization. The model performed well in various smear patterns which included background noises, overlapping cells, irregular stains and variational cell size. The YOLOv11 model can identify WBCs despite difficulties. This makes YOLO reliable for accurately identifying WBCs with lower error rate. Sample results are seen in Figures 4 and 5.

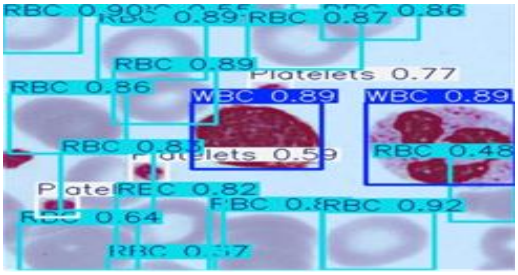


Figure 5. YOLOv11-Based WBC bounded box

5.2 Blast Classification and White Blood Cell Type Classification

The WBCATT dataset which includes single WBC cell images with their morphological attributes and classes was used to train ViT-Base-Patch16-224 multitasking model for determining different kinds of WBCs then separate blast from non-blast cells. The identification of blast is essential for the detection of leukemia, the model demonstrated good accuracy, precision, F1 and recall. The model identified well the Basophils, Eosinophils, Lymphocytes, Monocytes, and Neutrophils. The multitask learning strategy helped the model to better distinguish between similar cell types, and it reduced the chances of errors being committed by models trained for single tasks. The system provides a detailed diagnostic report of WBC class, Blast-status and reason for blast in the leukemia diagnosis. Example sample results are shown in Figures 6 and 7.

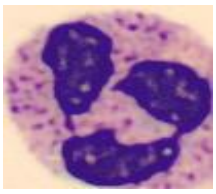


Figure 6. Cropped WBC 1 from YOLO sent to ViT for Blast prediction

The Figure 6 WBC sample was classified as a Neutrophil with a Normal blast status. The model predicted no blast-like morphological attributes across any of the 11 annotated features, confirming the cell as a mature, differentiated WBC with no diagnostic indicators of leukemia.

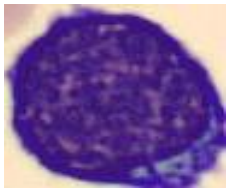


Figure 7. Cropped WBC 2 from YOLO sent to ViT for Blast prediction

The Figure 7 WBC sample was classified as a Monocyte with a Blast-Like status. The model flagged this cell based on four predicted morphological attributes: an unsegmented round nucleus shape, a high nuclear-to-cytoplasmic ratio, reduced or absent granules, and large cell size collectively meeting the threshold criteria for blast-like identification and warranting further clinical evaluation. Diagnostic report of WBC 2 of its class, Blast-status and reason for blast-like findings are provided.

5.3 **Quantitative Performance Summary**

The proposed YOLO-vision transformer framework is evaluated using standard detection and classification metrics along with mean Average Precision (mAP) are used to evaluate the WBC detection on the TXL-PBC dataset, are shown in Table 2. While the evaluation on the classification based on the WBCATT dataset, are shown in Table 3.

Table 2. WBC Detection Performance (TXL- PBC)

Class	Precision (%)	Recall (%)	mAP@0.50 (%)	mAP@0.50:0.95 (%)
WBC	98	99	99	90
RBC	97	94	99	92
Platelets	97	88	97	79
Overall	98	94	98	87

Table 3. WBC Classification Performance (WBCATT Dataset)

WBC Class	Precision (%)	Recall (%)	F1-Score (%)	Support (%)
Overall Accuracy	-	-	99	3099
Macro Average	99	99	99	3099
Weighted Average	99	99	99	3099

6 CONCLUSION

This research work proposes a fully automated, end-to-end framework for the detection of leukemia blast cells from blood smear images. The proposed model employs YOLOv11 for WBC localization from a blood smear sample vision transformer for multitask learning to predict the blast cells. This reduces manual workload of leukemia diagnosis. Experimental results on TXL-PBC dataset showed promising results, with achieved Overall Precision of 98%, Overall Recall of 94%, mAP@0.50 equal to 98%, and mAP@0.50:0.95 equal to 87%. These results confirm the ability of our model to locate White Blood Cells (WBC) in a smear sample accurately, which is critical for further classification. For classification on the WBCATT dataset, we obtained a collective accuracy of 99% and F1-scores of 99%, which is a strong result for discriminating WBCs subtypes and identifying blast and non-blast cells. Our approach provides a practical solution for large-scale screening of leukemia that will significantly reduce the pathologist's workload and allow for faster and more accurate diagnosis in the future. It would be helpful to compare the results with state-of-the-art CNN architectures like ResNet [11] and EfficientNet to identify which aspects of our model contribute to its high performance. Another interesting direction is to develop a multimodal deep learning framework that integrates different data modalities, such as visual, text, and tabular data. This would allow building more robust and generalizable models for healthcare tasks. Apart from that, the method can be extended to other blood-related diseases beyond leukemia. Finally, we need to evaluate the approach on larger-scale datasets and make it more clinically applicable by validating it on real-world data while also providing Grad-CAM and attention map explanations so that pathologists can understand and trust the algorithm's decisions.

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* denotes corresponding authors